FC7xxx DIO Integration Manual

Rev A0

Revision History

Revision	Date	Changes
A0	2025/01/14	Initial release

Table of Contents

Chapter 1 Introduction

1.1 Introduction

This integration manual describes the integration requirements for the DIO module.

Chapter 2 Building

2.1 Dependencies on Other Modules

Module configuration dependency

- Mcu: This module provides basic microcontroller initialization, and clock initialization for DIO module.
- Port: This module provides the port configurations of Port pins before they are used for DIO module.
- Common: This module is the basic module which used to choose the chip.

Module build dependency

- Common: This module provides common type for other modules.
- Rte: This module provides APIs to protect/unprotect some parts of code from Exclusive Areas.
- Det: This module is necessary for enabling Development error detection.

Module initialization dependency

- Mcu: To use the DIO module, the user needs to initialize the MCU module first.
- Port: To use the DIO module, the user needs to initialize PORT module first.

2.2 Files Required for Compile

DIO module files:

- _MCAL/Src/Dio/Src/Dio.c
- _MCAL/Src/Dio/Src/Dio_Hw.c
- _MCAL/Src/Dio/include/Dio.h
- _MCAL/Src/Dio/include/Dio_Hw.h
- _MCAL/Src/Dio/include/Dio_Hw_Types.h
- _MCAL/Src/Rte/include/Dio_MemMap.h
- _MCAL/Src/Dio/include/Dio_Reg.h
- _MCAL/Src/Dio/include/Dio_RegOps.h
- _MCAL/Src/Dio/include/Dio_Version.h
- _MCAL_generate/src/Dio_Cfg.c
- _MCAL_generate/include/Dio_Cfg.h

Common module files:

- _MCAL/Src/Common/include/Mcal.h
- _MCAL/Src/Common/include/Std_Types.h
- _MCAL/Src/Common/include/Platform_Types.h
- _MCAL/Src/Common/include/Compiler.h
- _MCAL/Src/Common/include/Compiler_Cfg.h
- _MCAL/Src/Common/include/CompilerDefinition.h

Det module files:

- Det.h

Rte module files:

- SchM_Dio.h

2.3 Add Plug-ins

DIO module plug-ins are developed for EB tresos Studio, so, to use DIO plug-ins on the EB tresos Studio, the user needs to add the DIO module to the EB plug-ins folder first.

1) Copy the DIO module(_MCAL/EB_Plugins/eclipse/plugins/Dio) folder to EB tresos plug-ins (EB/tresos/plugins/) folder.
2) Set the DIO module output location folder for the generated source file and header files.
3) Use EB tresos to configure the DIO module and generate source and header files.

Chapter 3 Memory

3.1 Sections in Memory Map

Section Name	Section Type	Description
DIO_START_SEC_CONFIG_DATA_8 DIO_STOP_SEC_CONFIG_DATA_8 DIO_START_SEC_CONFIG_DATA_16 DIO_STOP_SEC_CONFIG_DATA_16 DIO_START_SEC_CONFIG_DATA_32 DIO_STOP_SEC_CONFIG_DATA_32	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code(data).
DIO_START_SEC_CONFIG_DATA_UNSPECIFIED DIO_STOP_SEC_CONFIG_DATA_UNSPECIFIED	Configuration Data	Start and stop of Memory Section for Config Data.
DIO_START_SEC_CONST_BOOLEAN DIO_STOP_SEC_CONST_BOOLEAN DIO_START_SEC_CONST_8 DIO_STOP_SEC_CONST_8 DIO_START_SEC_CONST_16 DIO_STOP_SEC_CONST_16 DIO_START_SEC_CONST_32 DIO_STOP_SEC_CONST_32 DIO_START_SEC_CONST_ UNSPECIFIED DIO_STOP_SEC_CONST_ UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit or boolean. These variables are read only (rodata).
DIO_START_SEC_CODE DIO_STOP_SEC_CODE DIO_START_SEC_ RAMCODE DIO_STOP_SEC_ RAMCODE DIO_START_SEC_CODE_AC DIO_STOP_SEC_CODE_AC	Code	Start and stop of memory Section for Code(text).
DIO_START_SEC_VAR_NO_INIT_BOOLEAN DIO_STOP_SEC_VAR_NO_INIT_BOOLEAN DIO_START_SEC_VAR_NO_INIT_8 DIO_STOP_SEC_VAR_NO_INIT_8 DIO_START_SEC_VAR_NO_INIT_16 DIO_STOP_SEC_VAR_NO_INIT_16 DIO_START_SEC_VAR_NO_INIT_32 DIO_STOP_SEC_VAR_NO_INIT_32 DIO_START_SEC_VAR_NO_INIT_ UNSPECIFIED DIO_STOP_SEC_VAR_NO_INIT_ UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are never cleared and never initialized by startup code (bss).
DIO_START_SEC_VAR_ INIT_BOOLEAN DIO_STOP_SEC_VAR_ INIT_BOOLEAN DIO_START_SEC_VAR_ INIT_8 DIO_STOP_SEC_VAR_ INIT_8 DIO_START_SEC_VAR _INIT_16 DIO_STOP_SEC_VAR _INIT_16 DIO_START_SEC_VAR _INIT_32 DIO_STOP_SEC_VAR _INIT_32	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code (data).

Section Name	Section Type	Description
DIO_START_SEC_VAR _INIT_ UNSPECIFIED DIO_STOP_SEC_VAR _INIT_ UNSPECIFIED		

Chapter 4 Exclusive Area

DIO module using the services of Schedule Manger (SchM) for entering and exiting critical regions.

The following critical regions are used in the DIO driver:

Chapter 5 Interrupt Service Routine (ISR)

None

Chapter 6 Error Report

6.1 Det

Function Name	Error Type
Dio_ReadChannel	DIO_E_PARAM_INVALID_CHANNEL_ID
Dio_WriteChannel	DIO_E_PARAM_INVALID_CHANNEL_ID DIO_E_PARAM_LEVEL
Dio_FlipChannel	DIO_E_PARAM_INVALID_CHANNEL_ID
Dio_ReadPort	DIO_E_PARAM_INVALID_PORT_ID
Dio_WritePort	DIO_E_PARAM_INVALID_PORT_ID
Dio_MaskedWritePort	DIO_E_PARAM_INVALID_PORT_ID
Dio_ReadChannelGroup	DIO_E_PARAM_INVALID_GROUP_ID
Dio_WriteChannelGroup	DIO_E_PARAM_INVALID_GROUP_ID
Dio_GetVersionInfo	DIO_E_PARAM_POINTER

6.2 Dem

None

Chapter 7 Function Calls

7.1 Function Calls during Startup

None

7.2 Function Calls during Shutdown

None

7.3 Function Calls during Wake-up

None

7.4 Function Calls during Runtime

None

Chapter 8 Other Requirements

8.1 Notification, Callback, Callout

None

8.2 Macros

None

Chapter 9 Integration Steps

1) Configure the DIO module and generate configuration files (please refer to Building chapter for details).
2) Configure appropriate memory sections in linker file or other (please refer to Memory chapter for details).
3) Build the DIO module with dependent modules.